ICS 6B F26 Take Home Exam 5

Due: October 30, 2026 at 11:59PM

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UCI NetID :

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(**alpha-numeric; NOT your student ID**)

- Read the instructions of each question carefully.
- Show your work. Where applicable, partial credit will be given on the work shown.
- Please keep in mind the academic honesty guidelines. This take-home exam is to be completed individually, with no outside help. You may use any resources from our class (ZyBooks and resources from Canvas), but you may not use any other online resources.
- You may choose to print the exam or use a digital editor for completing the exam. It is required that you use this PDF to complete your work. If you have no access to a printer or digital tools to fulfill the exam, feel free to reach out to the staffs regarding your concerns.
- If you have any questions, please post a private Ed or attend available Office Hours. Note that we are not allowed to provide specific help to answering the exam questions.
- There are **5** problems on this test.
- There are a total **18** of points.
- Good luck!

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Problem 1 (5 point)

For each of the following functions choose ***all*** valid options that describe the function f.

1. $f : \mathbb{Z} \to \mathbb{Z}$ such that $f(x) = 3x - 1$.
   ☐ one-to-one ☐ not one-to-one ☐ onto ☑ not onto ☐ bijection
2. f is inverse of g, such that $g : \{1, 3, 5, 7, 9\} \to \{0, 1, 2, 3, 4\}$ and $g(x) = \frac{x-1}{2}$.
   ☐ one-to-one ☐ not one-to-one ☐ onto ☐ not onto ☐ bijection
3. $f : \mathbb{Z} \to \{0, 1, 2, \ldots, n-1\}$ such that $f(x) = ((x + 2) \mod n)$. $a \mod b$ returns the remainder when we divide a by b. For example $(11 \mod 3) = 2$.
   ☐ one-to-one ☐ not one-to-one ☐ onto ☐ not onto ☐ bijection
4. $f = g \circ h$ where $g : \mathbb{Z} \to \mathbb{Z}$ such that $g(x) = 2x+3$ and $h : \mathbb{Z} \to \mathbb{Z}$ such that $h(x) = x^2+1$.
   ☐ one-to-one ☐ not one-to-one ☐ onto ☐ not onto ☐ bijection
5. $f : \mathbb{R} \to \mathbb{Z}$ such that $f(x) = \lfloor x^3 \rfloor$.
   ☐ one-to-one ☐ not one-to-one ☐ onto ☐ not onto ☐ bijection

Problem 2 (3 points)

Consider a function $f : \mathcal{P}(\mathbb{A}) \to \{0, 1\}^n$, where $\mathbb{A} = \{a_1, a_2, \ldots, a_n\}$ is a set with n elements. f maps subsets of $\mathbb{A}$ to binary strings of length n.

$$f(X) = y \text{ where } \begin{cases} y_i = 1 & \text{if } a_i \in X \\ y_i = 0 & \text{if } a_i \notin X \end{cases} \tag{1}$$

where y_i represents the ith digit of string y. f is a bijection. Write "DNE" if the function is not well defined.
If $\mathbb{A} = \{10, 0, 1, 100, 11\}$:

1. What is $f(\{0, 100, 11\})$? 01011
2. What is $f^{-1}(10110)$? {10, 1, 100}
3. Choose all of the following which are equal to $f^{-1}(f(\{1, 10, 11\}))$.
   ☐ 10101 ☑ $\{11, 1, 10\}$ ☐ 11011 ☐ $\{0, 100\}$ ☐ $\mathbb{A}$ ☐ $\{0, 1\}^n$

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Problem 3 (4 points)

Consider sets $A = \{1, 2, 3, 4, 5, 6\}, B = \{1, 2, 3\}$ and $C = \{1, 2, 3, 4\}$. Let $F_{X,Y}$ represent the set of all functions from domain X to codomain Y. Select True or False for each of the following.

1. If $X = A \times B \times C$ and $Y = A \times A$ then $\exists f \in F_{X,Y}$ such that f is one-to-one.
   ◯ True ◯ False
2. If $X = A^2 \times \{0, 1\}^2$ and $Y = B^2 \times C^2$ then $\exists f \in F_{X,Y}$ such that f is onto.
   ◯ True ◯ False
3. If $X = \{1, 2\} \times B \times C$ and $Y = A \times C$ then $\exists f \in F_{X,Y}$ and $\exists g \in F_{Y,X}$ such that f is the inverse of g.
   ◯ True ◯ False
4. If $X = B^2$, $Y = A \times B$ and $Z = \{0, 1\}$ then $\forall f \in F_{X,Y}\ \exists g \in F_{Y,Z}$ such that $g(f((1, 2))) = g((6, 3)) = 1$.
   ◯ True ◯ False

Problem 4 (2 points)

Consider the domain A and codomain B. f is a map from A to B. Write a quantified expression using A, B (i.e. expressions of the form $\forall x \in A\ \exists y \in B \ldots$) and f that makes f a well-defined function. Recall a map is an arrow diagram from the domain to the codomain or more formally, $f \subseteq A \times B$.

$\forall x \in A\ \exists y \in B\ \forall z \in B\ (((x,y) \in f) \wedge ((x,y) \in f \wedge (x,z) \in f) \rightarrow y = z)$

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Problem 5 (4 points)

For each function write whether or not it has an inverse. If it does, write the inverse and explain why it is an inverse of the function. You may describe the inverse in plain English. If there is no inverse write 'No inverse' and you do not need to explain why there is no inverse.

1. The function $H : \{0,1\}^{2n} \to \{0,1\}^n$ for some $n \in \mathbb{Z}^+$: the i-th bit of the output of H is the $(2i-1)$-th bit of the input. For example (for $n = 3$) $H(100110) = 101$.

   Inverse (or simply 'No inverse'):

   No inverse

   Explanation:

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2. The function $W : \{0,1\}^n \to \{0,1\}^n$ for positive even n. W moves every digit in an even place to the next even place i.e. for the $i-th$ bit if i is even then put the bit in $i+2$-th spot and so on til the end where the n-th bit is put into the 2nd bit's vacancy, else if i is odd then it flips the bit. For example, $W(111000) = 000110$.

Inverse (or simply 'No inverse'):

For all odd position i, set i-th bit to 1 - yi
For every even position i, set the i-th bit to yi + 2, where the even positions wrap around cyclically

Explanation:

W flips each odd-position bit, and flipping is its own inverse.
It also cyclically shifts the even-position bits by two places, so the inverse shifts them back by two places.